Nicolas Bigeard

MACHINE LEARNING RESEARCHER (MSC CANDIDATE)

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Paris, France (relocating for the placement)

SUMMARY

MSc candidate in Financial Engineering & Quantitative Finance at ESILV (diplôme d'ingénieur, expected 2027) and research collaborator at Centre Borelli, ENS Paris-Saclay. Sole author of a **NeurIPS 2026 AI for Science Workshop** paper (poster) and of a **matching-minimax-rates** preprint; first author of Tucker-CAM (causal discovery scaling to 2,889 variables). My work asks what searching for structure actually costs, across theorem proving, recurrent-depth models, diffusion operators and causal discovery. Seeking a research placement starting February or March 2027.

CORE COMPETENCIES

Machine Learning Research | Statistical Learning Theory | Minimax Rates | Causal Discovery | Deep Learning / Transformers | Reasoning in LLMs | Lean 4 / Theorem Proving | Large-scale Data Pipelines

RESEARCH EXPERIENCE

Centre Borelli, ENS Paris-Saclay — May 2026 - Present

Research Intern, continuing as collaborator (with Argyris Kalogeratos)

Paris, France

- Derived the first out-of-sample extension for multi-view diffusion operators: proved every operator in the family factors as W = Σ_v P_v S_v with S_v independent of the test rows, so extension is exact on the training set and training-free.
- Built the family's first inductive benchmark (19 datasets, 10 seeds, bootstrap intervals, exact sign tests, Holm correction over 17 comparisons): +0.096 to +0.108 inductive AMI over five published competitors; matches SpecRaGE (TMLR 2025) at half the compute.
- Reported negative results as such: the equal-weight constant trajectory ties the searched operator (0.413 vs 0.417) and wins at six of nine noise levels; an Eckart-Young argument caps the learned-encoder alternative at the truncated SVD, at roughly 300× the cost.

SauronID — Feb. 2026 - Present

Co-founder & CEO

- Fail-closed authorization and verifiable-audit boundary for AI agents (self-hostable gateway and TypeScript SDK; tamper-evident receipts). sauronid.eu

Independent research project, supervised by Yiannis Vlassopoulos (Athena Research Center; visiting IHES) — Jan. 2026 - Mar. 2026

Game-theoretic interpretability of chess engines and neural networks

Remote / Paris, France

Paul Wurth S.A. — Apr. 2025 - Feb. 2026

Research Intern, Causal Discovery (Machine Learning)

Luxembourg - supervised by Cédric Schockaert, Head of Data Science

- Built Tucker-CAM, a non-linear causal discovery pipeline for high-dimensional industrial time series (P-spline mechanisms, Tucker-factorised coefficient tensor): parameters $O(d^2)$ to $O(dr)$, scaling to $d = 2{,}889$ variables.
- State-aware anomaly scoring plus graph traversal for root-cause analysis: Range-F1 0.9977 and AUC-PR 0.9782 on NASA Telemanom.
- Worked with significant autonomy: extended DynoTEARS with a rolling-window scheme, integrated Fast CAM and Tucker decomposition, formulated the supporting mathematical proofs and handled the entire software implementation. First author of the resulting preprint.

PUBLICATIONS & PROJECTS

[P2] Measure the Library First - NeurIPS 2026 AI for Science Workshop (poster), sole author

Audited my own arXiv-to-Lean 4 pipeline: of twenty theorems reported as verified, one carries mathematical content. Traced six paths to "proved" without a proof. On miniF2F, twenty Mathlib tactics with no model close 36.1% vs 26.0 ± 0.9% for a retrieval-augmented search loop (search advantage ≤ 2.1 points at 95%).

doi:10.5281/zenodo.21937508

[P1] Two Costs of Tying Depth: Matching Minimax Rates - sole author

Proved matching minimax bounds (Assouad construction for the continuous cores, Fano construction over a packing of assignments) for depth-tied students; regime threshold $P_\star = L \log m / m$, exceeded by seven orders of magnitude by whole-block tying. Real-model study over nine pretrained teachers (GPT-2, Pythia), reported as mixed.

doi:10.5281/zenodo.23047408

[P3] Tucker-CAM - first author, with C. Schockaert

Tractable non-linear causal discovery for high-dimensional industrial time series anomaly detection; Range-F1 0.9977, AUC-PR 0.9782 on NASA Telemanom.

doi:10.5281/zenodo.19888858 | github.com/tejoker/Unaite_summer_research

[P5] Nash Initialization for Recurrent-Depth Transformers - preprint

LayerNorm applies a per-cycle rank-(d−2) projection; O(1/T) residual scaling is the critical exponent for stability without normalisation.

doi:10.5281/zenodo.20021200

US Equities Tick-to-Candle Pipeline and Return Model - sole author

4-stage backpressure pipeline turning tick-level trades and quotes into 1-minute OHLCV Parquet, 2005-2025, about 10,500 tickers per day; external sort plus streaming holds RAM under 10 GB on 35 GB inputs, about 10 min per trading day.

Python | github.com/tejoker/PipelineStocks

EDUCATION

MSc / Diplôme d'ingénieur, Financial Engineering & Quantitative Finance - ESILV, Paris, France — Sept. 2023 - 2027

Five-year integrated French engineering degree entered in Year 3 after CPGE. Year 3: Félicitations du jury, 60/60 ECTS. Year 4: research modules 17.33 and 18.41/20. 2025-26: approved research gap year. Coursework: Advanced Probability, Advanced Statistics, Stochastic Processes, Machine Learning. Expected graduation: 2027.

Classes Préparatoires (CPGE) MPSI / MP - Lycée Saint François d'Assise, La Roche-sur-Yon — 2021 - 2023

Ranked 1st of 24 in MP in both semesters; 1st in mathematics (MP, semester 2).

CERTIFICATIONS

AMF professional certification; Bloomberg Market Concepts; Bloomberg Finance Fundamentals

AWARDS

AI hackathons: Agent Hackathon (Hugging Face, Anthropic); Iterate Paris (Anthropic)

RESEARCH INTERESTS

AI for mathematics (reasoning rather than formalisation), statistical learning theory, causal discovery, spectral methods.

SKILLS

Programming: Python, C++, Lean 4, OCaml, SQL

Libraries: PyTorch, CUDA, NumPy, Polars, PyArrow, scikit-learn, Docker, Linux, Git, LaTeX

Mathematics: minimax theory, model selection and oracle inequalities, spectral and diffusion geometry, convex optimisation

Methods: causal structure learning, tensor decomposition, time-series anomaly detection, Monte-Carlo tree search, RL for reasoning, experimental design with multiple-comparison correction

Languages: French (native), English (fluent), Spanish (B1)